6.2. SVOJSTVA SUSTAVA

Bezmemorijski sustavi

$\forall t \in \mathbb{R} \quad y(t) = S(u(t)) = S(u)(t)$

$\forall n \in \mathbb{Z} \quad y[n] = S(u[n]) = S[u][n]$

nema operacija, kašnjenja „tu i sada"

$y(t) = \frac{1}{2} u(t)$

Sustav sa beskonačnom memorijom

$\forall t \in \mathbb{R} \quad y(t) = S(u\langle -\infty, t], (t))$

$\forall n \in \mathbb{Z} \quad y[n] = S(u\langle -\infty, n], [n])$

koliki je izlaz našeg sustava od početka gledanja do sada

$y(t_0, t) \qquad u(t_0, t)$

i koliki je ulaz od tada do sada

$\langle -\infty, t_0]$ označimo jednim brojem = poč. stanje sustava

$y(t) = S(\alpha_i, u(t_0, t))(t)$

$y[n] = S(\alpha_i, u[n_0, n])[n]$

poč. stanja sustava

odziv mirnog sustava

$$y(t) = \int_{-\infty}^{t} u(\tau)d\tau = \underbrace{\int_{-\infty}^{t_0} u(\tau)d\tau}_{\text{poč uvjet „prije jučer"}} + \underbrace{\int_{t_0}^{t} u(\tau)d\tau}_{\text{„jučer i danas"}} = y(t_0) + \int_{t_0}^{t} u(\tau)d\tau$$

od poč. do sada

ako je $y(t_0) = 0$ onda je odziv mirnog sustava

u disk. svijetu:

$$y[n] = \sum_{m=-\infty}^{n} u[m] = \underbrace{\sum_{m=-\infty}^{n_0-1} u[m]}_{\text{poč. stanje}} + \sum_{m=n_0}^{n} u[m]$$

Primjer: kako provjeriti

Sustav za usrednjavanje $y[n] = \frac{1}{L+1} \sum_{m=0}^{L} u[n-m]$

$$y[n] = \frac{1}{4} \sum_{m=0}^{3} u[n-m] = \frac{1}{4} \left(u[n] + u[n-1] + u[n-2] + u[n-3] \right)$$

→ sustav ima memoriju jer trebamo prošlost

→ ako se vrši neka operacija nad u tj da je unutar u[...] modificira onda ima memoriju

U Matlabu: video 33. → 18:31

Npr. popijemo pivo → puščamo promile

popijemo 2. pivo → napuščamo više promila ↳ tijelo pamti

Kauzalnost i nekauzalnost SUSTAVA!

Kauzalni

→ ako izlaz sustava u sadašnjosti ovisi samo o sadašnjosti ili prošlosti

$\forall t \in \mathbb{R} \quad y(t) = S(u_{<-\infty,t]})(t)$

$\forall n \in \mathbb{Z} \quad y[n] = S(u_{<-\infty,n]})[n]$

Nekauzalan ako ovisi samo o budućnosti

→ izlaz ovisi o ulazu u prošlosti, ali može ovisiti i o budućnosti

$\forall t \in \mathbb{R} \quad y(t) = S(u_{<-\infty,t_0]})(t)$

$\forall n \in \mathbb{R} \quad y[n] = S[u_{<-\infty,n_0]}][t]$

$n < n_0 < \infty$

- vozimo, vidimo auto i kočimo jer vidimo da ćemo se u budućnosti sudariti

Primjer: $y[n] = u[n+2]$

$y[4] = u[6]$

za izlaz u trenutku 4 moramo znati ulaz u trenu 6. budućnosti

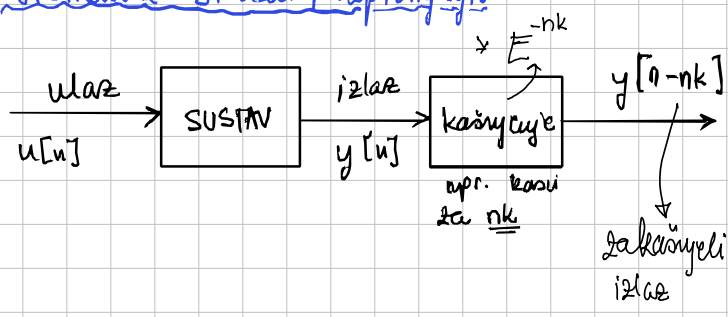
→ za sadašnjost trebamo ulaz iz budućnosti

← NEKAUZALAN

Vremenska stalnost / promjenjivost

→ ako zakasnimo ulaze, hoće li se jednako zakasniti i izlazi

Vremenski stalan / nepromjenjiv

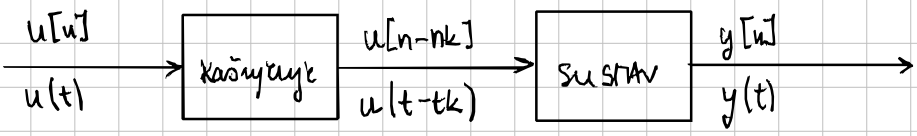


* diskretni

$\boxed{E^{-nk}}$

* vremenski:

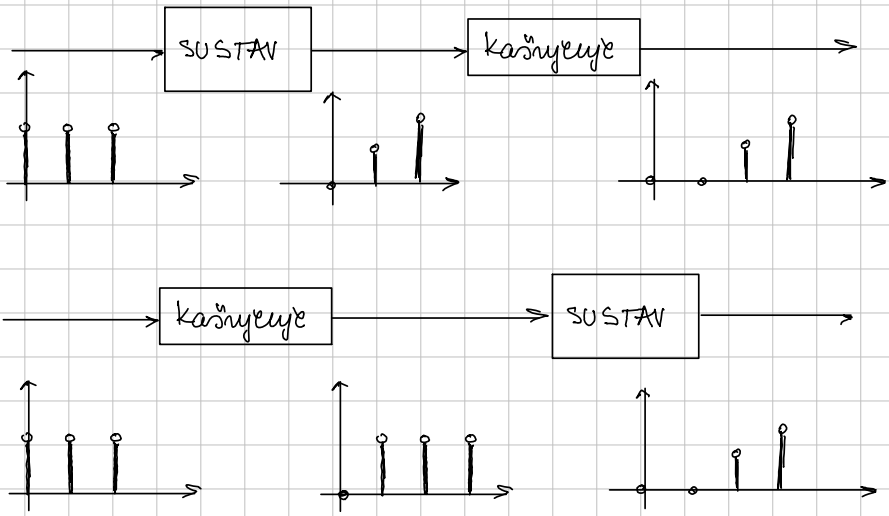
$\boxed{E^{-1}_{tk}}$



Ako vrijedi:

$$\left.\begin{aligned} y[n-nk] &= y_1[n] \\ y(t-tk) &= y_1(t) \end{aligned}\right\} \text{ vremenski stalan ✓}$$

Primjer stalnosti



budući da su jednaki

$\Downarrow$

vrijedi vremenska stalnost

Primjer 2.: $y[n] = \frac{1}{L+1} \sum_{m=0}^{L} u[n-m]$

$$u[n] \xrightarrow{S} y[n] = \frac{1}{L+1} \sum_{m=0}^{L} u[n-m] \xrightarrow{E^{-nk}} y[n-nk] = \frac{1}{L+1} \sum_{m=0}^{L} u[n-nk-m]$$

$$u[n] \xrightarrow{E^{-nk}} \underbrace{u[n-nk]}_{u_1[n]} \xrightarrow{S} y_1[n] = \frac{1}{L+1} \sum_{m=0}^{L} u_1[n-m]$$

$$y_1[n] = \frac{1}{L+1} \sum_{m=0}^{L} u[n-nk-m]$$

$\Downarrow$

Primjer: kontinuirani $y(t) = t \cdot u(t) \cdot 4$

$\rightarrow$ nema memoriju

$\rightarrow$ kauzalan

$$u(t) \xrightarrow{S} y(t) = 4t \cdot u(t) \xrightarrow{E^{-1}_{tk}} y(t-tk) = 4(t-tk) \cdot u(t-tk)$$

$$u(t) \xrightarrow{E^{-1}_{tk}} u(t-tk) = u_1(t) \xrightarrow{S} y_1(t) = 4t \cdot u(t)$$

$\times$ nije stalan

$$y_1(t) = 4 \cdot t \cdot u(t-tk)$$

$y(t-tk) \neq y_1(t)$ vremenski promjenjiv

Simulink 25:30, video 34

Svojstvo linearnosti

ako ima pri uvjet → NIJE LIN!

→ mora biti homogen i aditivan

Homogenost $u_1 \longrightarrow \alpha u_1 \xrightarrow{S} S(\alpha u_1) = \alpha S(u_1) = \alpha y_1$

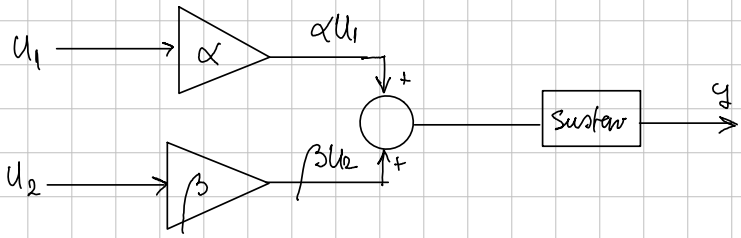
ako je uvjet zadovoljen → homogen ✓

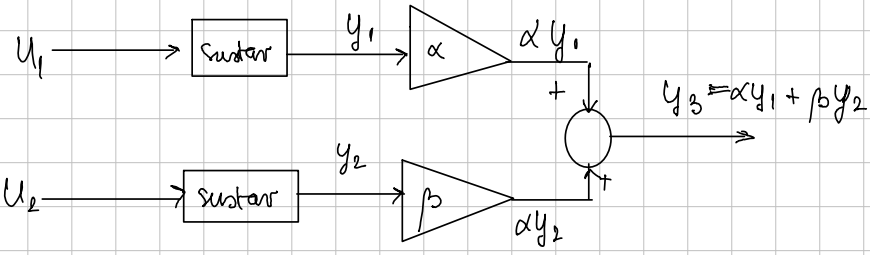
- koliko puta se poveća ulaz toliko puta će i izlaz

Aditivnost $u_1 + u_2 \xrightarrow{S} S(u_1+u_2) = S(u_1) + S(u_2) = y_1 + y_2$ ✓

- ako zbrojimo ulaze i to ubacimo u sustav, rez mora biti kao da smo ih odvojeno ubacili u sustav i tek onda zbrojili

Superpozicija $S(\alpha u_1 + \beta u_2) = \alpha S(u_1) + \beta (S u_2)$ → linearan





$y_1 = y_3$ ✓

LINEARAN

Primjer: $y(t) = 4t \cdot u(t)$

$$\left.\begin{array}{l} u_1 \longrightarrow \alpha u_1(t) \\ u_2(t) \longrightarrow \beta u_2(t) \end{array}\right\} + \quad u(t) = \alpha u_1(t) + \beta u_2(t)$$

$$\rightarrow y(t) = 4t \cdot (\alpha u_1(t) + \beta u_2(t))$$

$$y(t) = 4t\alpha u_1(t) + 4t\beta u_2(t)$$

$$\left.\begin{array}{l} u_1 \longrightarrow y_1(t) = 4t\, u_1(t) \longrightarrow \alpha y_1(t) = \alpha 4t \cdot u_1(t) \\ u_2 \longrightarrow y_2(t) = 4t\, u_2(t) \longrightarrow \beta y_2(t) = \beta \cdot 4t \cdot u_2(t) \end{array}\right\} + \quad y(t) = 4t\alpha u_1(t) + 4t\beta u_2(t)$$

✓

LINEARAN

Primjer 2) $y(t) = \sin(u(t))$

$$\left.\begin{array}{l} u_1(t) \longrightarrow \alpha u_1(t) \\ u_2(t) \longrightarrow \beta u_2(t) \end{array}\right\} + \quad \alpha u_1(t) + \beta u_2(t) = u(t)$$

$$\Rightarrow \underline{y(t) = \sin(\alpha u_1(t) + \beta u_2(t))}$$

* ponekad moramo još poznati trigonometrijske identitete *

$$\left.\begin{array}{l} u_1(t) \longrightarrow y_1(t) = \sin(u_1(t)) \longrightarrow \alpha y_1(t) = \alpha \sin(u_1(t)) \\ u_2(t) \longrightarrow y_2(t) = \sin(u_2(t)) \longrightarrow \beta y_2(t) = \beta \sin(u_2(t)) \end{array}\right\} + \quad \underline{y_3 = \alpha \sin(u_1(t)) + \beta \sin(u_2(t))}$$

$y(t) \neq y_3(t) \longrightarrow$ nije linearan

Matlab provjera linearnosti = 13:45 (SIS 35.)

* integrator je sustav sa memorijom jer $[-\infty, t]$

Sustav integratora $\quad y(t) = \int_{-\infty}^{t} u(\tau)\, d\tau \quad \rightarrow$ je li sustav linearan

$$\left.\begin{array}{l} u_1(t) \longrightarrow \alpha u_1(t) \\ u_2(t) \longrightarrow \beta u_2(t) \end{array}\right\} + \Rightarrow u(t) = \alpha u_1(t) + \beta u_2(t) \xrightarrow{S} y(t) = \int_{-\infty}^{t} u(\tau)\, d\tau$$

konstante

$$y(t) = \int_{-\infty}^{t} \alpha u_1(\tau) + \beta u_2(\tau)\, d\tau = \underline{\int_{-\infty}^{t} \alpha u_1(\tau)\, d\tau + \int_{-\infty}^{t} \beta u_2(\tau)\, d\tau}$$

$$\left.\begin{array}{l} u_1(t) \xrightarrow{S} y_1(t) = \int_{-\infty}^{t} u_1(\tau)\, d\tau \longrightarrow \alpha y_1(t) \\ u_2(t) \xrightarrow{S} y_2(t) = \int_{-\infty}^{t} u_2(\tau)\, d\tau \longrightarrow \beta y_2(t) \end{array}\right\} + \Rightarrow \alpha y_1(t) + \beta y_2(t)$$

$$= \underbrace{\alpha \int_{-\infty}^{t} u_1(\tau)\, d\tau + \beta \int_{-\infty}^{t} u_2(\tau)\, d\tau}_{\text{isti su}}$$

$\Rightarrow$ vrijedi linearnost

* $y(t) = \int_{-\infty}^{t} u(\tau)\, d\tau = \underbrace{y(t_0)}_{\text{„davno"}} + \underbrace{\int_{t_0}^{t} u(\tau)\, d\tau}_{\text{„nedavno"}}$

$$\left.\begin{array}{l} u_1(t) \rightarrow \alpha u_1(t) \\ u_2(t) \rightarrow \beta u_2(t) \end{array}\right\} u(t) = \alpha u_1(t) + \beta u_2(t) \longrightarrow \underline{y(t) = y(t_0) + \int_{t_0}^{t} (\alpha u_1(\tau) + \beta u_2(\tau))\, d\tau}$$

$$\left.\begin{array}{l} u_1(t) \rightarrow y_1(t) = y(t_0) + \int_{t_0}^{t} u_1(\tau)\, d\tau \longrightarrow \alpha y_1(t) = \alpha y(t_0) + \alpha \int_{t_0}^{t} u_1(\tau)\, d\tau \\ u_2(t) \rightarrow y_2(t) = y(t_0) + \int_{t_0}^{t} u_2(\tau)\, d\tau \longrightarrow \beta y_2(t) = \beta y(t_0) + \beta \int_{t_0}^{t} u_2(\tau)\, d\tau \end{array}\right\} +$$

poč. uvjeti se ne poklapaju!

$y(t) \neq y_3(t) \longrightarrow$ sustav nije linearan ZBOG POČ. UVJETA!!

$$y(t) = \underbrace{y(t_0)}_{\text{zbog njega je sustav nelinearan}} + \underbrace{\int_{t_0}^{t} u(\tau)\,d\tau}_{\text{odziv mirnog sustava}}$$

INKREMENTALNO LIN. SUSTAV

→ ako nema ($y(t_0)=0$) ⇒ odziv nepobuđenog sustava

Ako je na ulazu 0, na izlazu mora biti 0 da bi bila ljv

OBRAT NE VRIJEDI (NUŽNO)

BIBO STABILNOST „bounded input bounded output"

- ako smo na ograničenu pobudu dobili ograničeni odziv → sustav stabilan

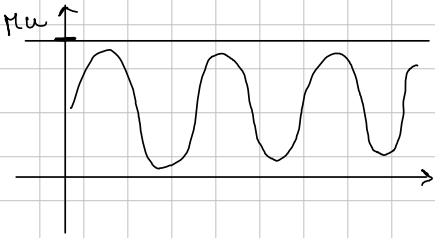
$$|u(t)| \leq M_u < \infty \implies |y(t)| \leq M_y < \infty$$

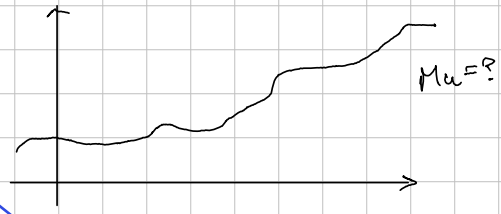
(ograničeni ulaz) stabilan

$$|u[n]| \leq M_u < \infty \implies |y[n]| \leq M_y < \infty$$

Primjer: $y(t) = 7u(t) + 6$

$|u(t)| \leq M_u < \infty \longrightarrow y(t) = 7M_u + 6$ ✓ stabilan

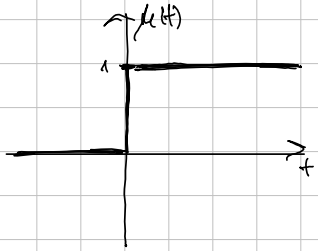




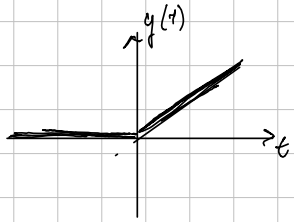
Primjer 2.)* Možemo promjeravati stabilnost ili pogodimo/znamo neku vrijednost za koju je ∞ ⟹ znamo da nije stabilno

$y(t) = \int_{-\infty}^{t} u(\tau)\,d\tau$ npr. za $u(\tau) = \tau$ sigurno ide u ∞

$$u(t) = \mu(t) \implies y(t) = \int_{-\infty}^{t} \mu(\tau)\,d\tau = \int_{0}^{t} 1\,d\tau = \tau\Big|_0^t = t$$



⟹



NIJE STABILAN

Primjer: Diskretan sustav za uprosječavanje

$$y[n] = \frac{1}{L+1} \sum_{m=0}^{L} u[n-m]$$

$$|u[n]| \leq M_u < \infty \longrightarrow |y[n]| \leq \frac{1}{L+1} \sum_{m=0}^{L} M_u = \frac{\cancel{(L+1)} M_u}{\cancel{L+1}} = M_u < \infty$$

izlaz nije veći od M_u → BIBO STABILAN ✓

Provjera bibo stabilnosti u Matlabu: SIS 37 [5:20]